

## Problem Statement 2: Audio Tracks

### Problem Statement

A music analytics team tracks streaming performance and audio characteristics for top chart tracks, but has no consolidated view of which artists and songs are driving streams or what audio traits (danceability, energy, tempo) correlate with chart success. This makes it hard to brief playlist curators or A&R teams without manually cross-referencing spreadsheets.

**Objective:** Build a Power BI report that gives the analytics team a self-service view of streaming performance by artist and track, alongside audio characteristics, to support playlist curation and release strategy decisions.

### Key business questions

1. Which artists and tracks generate the most streams?
2. Do audio characteristics such as danceability, energy, or BPM differ meaningfully between high-stream and low-stream tracks?

**Scope:** Track-level chart data (track name, artist, release year/month/day, streams, BPM, key, mode, and audio-feature percentages); one Power BI report with an executive summary page, built using Power Query transformations and implicit (drag-and-drop) measures.

### Data Acquisition and Preparation

Import the provided CSV into Power BI Desktop (Home → Get Data → Text/CSV → Transform Data) and perform the following Power Query transformations:

| Step                      | Action                                                                                                                                             |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Change data type          | Combine released_year/released_month/released_day into a single Release Date column, and fix streams if it loads as text (convert to Whole Number) |
| Remove duplicates         | Check for and remove duplicate rows on track_name + artist(s)_name                                                                                 |
| Clean and trim            | Trim/Clean extra spaces from track_name and artist(s)_name                                                                                         |
| Replace null/blank values | Check key column for blanks; replace with 9                                                                                                        |

End with Close & Apply to load the cleaned data into the model.

## Interactive Report and Executive Summary

All figures use implicit measures — drag a field into a visual and set the aggregation in the Values well.

### Executive Summary KPI cards

- Total Tracks (Count of track\_name)
- Total Streams (Sum of streams)
- Average BPM (Average of tempo)
- Average Danceability % (Average of danceability in percentage)

### Exploratory Data Analysis

- Top 10 Artists by Streams count
- Track Count by Key or Mode (Major/Minor)
- Average Energy % or Danceability % by Mode
- Average Streams by release year

### Table graph

- Track Name, Artist, Streams, BPM, Danceability %, Energy % — sortable and filterable

### Tooltip

- Build a tooltip page showing upbeat\_score % and Released Date when hovering over a track/artist bar

### Insights

1. Which artists and tracks generate the most streams?
2. Do audio features such as danceability, energy, or BPM differ between high-stream and low-stream tracks?

End with 1–2 concrete recommendations for playlist curation or release strategy based on these findings.